

PhD Candidate Gustavo Togeiro de Alckmin

Education

- PhD candidate at Wageningen University (NL) and University of Tasmania (AUS) 2016 - 2020

Thesis by Publication: Determining Biophysical Attributes of Pasture Using Very High Resolution Imaging Techniques. *The four chapters should be published before or after thesis submission.*

- MSc. AgrisMundus – Montpellier SupAgro (FR) and Uni. Politecnica de Madrid (ES) 2012 – 2014

Thesis: Use of Remote Sensing Techniques to Enhance Forage Management: Unmanned Aerial Vehicle

- BSc. University of São Paulo – Luiz de Queiroz (BR) 2005 – 2010

Bachelor of Agriculture.

Exchange Programs as an undergraduate student

- Texas Tech University, Lubbock, TX, USA 2010
- Wageningen University, The Netherlands 2009

Professional Experiences:

- Wageningen University and University of Tasmania 2018 to 2020
Remote Sensing instructor at undergraduate and graduate levels
Instructor on Image Processing (ENVI), hyperspectral data collection and geospatial statistics
- AB Agro Imagem (start-up) Founder and Research Leader 2014 - 2015
Startup Company for UAV development and remote sensing techniques for forage management.
- Kansas State University www.kstate.edu 2014
Short-term Scholar
Development of a UAV and deployment of remote sensing techniques for forage assessment.
- Maison de la Télédétection <http://www.teledetection.fr/> 2013
Intern, Montpellier SupAgro. Remote Sensing Institute
Analysis and Evaluation of a Livestock Vegetation Index Based Insurance.
- Czarnikow Brasil LLC www.czarnikow.com 2010 - 2011
Commercial Executive (Sugar Trader)
Market Intelligence for Industrial Users, Risk Management, Futures Pricing Advisory, International Sugar Trading.
- ABC Foundation www.fundacaoabc.org.br 2009
Research and Development (Internship)
Applied research on phytopathology and entomology on wheat, soybean, beans and maize.
- Universidade de São Paulo <http://www.en.esalq.usp.br/> 2008
Teaching Assistant Laboratory of GIS

Extra-Curricular Experiences:

- YPARD-FAO – Country Representative (Brazil). 2015 - 2016
- Erasmus Mundus – Global Representative for the MSc. Program “AgrisMundus” 2012 - 2013
- Intern at Beef Production Unit at Castres, France. 2012
- Intern at Beef Production Units at Fortin del Patria, Lincoln, Buenos Aires, Argentina. 2008
- Volunteer as a handler at the horse-riding therapy program. Piracicaba. Brazil 2007
- Volunteer Teacher for underprivileged students. Campinas, Brazil 2004

Language Skills:

Portuguese (Fluent); Spanish (Fluent); French (Fluent) English: TOEFL – Internet-based score: (116/120)

Programming Languages:

R (Proficient); Python (Intermediate); HTML /WebDesign (Intermediate)

Teaching Experience:

- GIS, Remote Sensing and Spatial Statistics – Teacher Assistance. 2018-2020
- GIS and Remote Sensing (undergrad level) – Teacher Assistance 2008
- Math and Physics (high-school level) for university admission exams 2004

Awards

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| ▪ Goetz Instrument Award (ASD PanAnalytical) – | 2018/2019 |
| ▪ School of Land and Food Scholarship and Tuition (UTAS) - | 2016 |
| ▪ Msc. Thesis Development Grant – | 2014 |
| ▪ Erasmus Mundus Scholarship – Master of Sciences AgrisMundus | 2012 |
| ▪ Capes – FIPSE Scholarship - Consortium for Dryland Development | 2010 |

Academic Publications:

▪ Journal peer-reviewed:

Togeiro de Alckmin, G.; Kooistra, L.; Rawnsley, R.; de Bruin, S.; Lucieer, A. Retrieval of Hyperspectral Information from Multispectral Data for Perennial Ryegrass Biomass Estimation. *Sensors* 2020, 20, 7192. <https://doi.org/10.3390/s20247192>

Alckmin, G. T. , Lucieer, A., Roerink, G., Rawnsley, R., Hoving, I., & Kooistra, L. (2020). Retrieval of Crude Protein in Perennial Ryegrass Using Spectral Data at the Canopy Level. *Remote Sensing*, 12(18), 2958. <https://doi.org/10.3390/rs12182958>

Alckmin, G. T. , Kooistra, L., Rawnsley, R., & Lucieer, A. (2020). Comparing methods to estimate perennial ryegrass biomass: canopy height and spectral vegetation indices. *Precision Agriculture*, 1–21. <https://doi.org/10.1007/s11119-020-09737-z>

▪ Conference peer-reviewed:

Alckmin, G. T. , van der Merwe, D., José Antonio, M., & Tisseyre, B. (2019). Employing false color infrared cameras for biomass estimation on natural grassland. *In European Conference on Precision Agriculture (pp. 19–20)*.

Alckmin, G. T. , Kooistra L. Lucieer, A., & Rawnsley, R. (2019). Feature filtering and selection for dry matter estimation on perennial ryegrass: a case study of vegetation indices. *In The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences (Vol. 42)*.

▪ Conference :

Kooistra L. van der Beek L., Franceschini M., Bartholomeus H., **Alckmin G. T. ,** Berendonk C., Modelling the seasonal traits of production grasslands from UAV-based imaging spectroscopy (2019). *In 11th EARSeL Workshop on Imaging Spectroscopy, Brno, Czech Republic*.

Torres, R. V., Dias, L. F. R. S., **Alckmin, G. T. ,** Fiorio, P. R., & Angulo Filho, R. (2007). Influência da cobertura vegetal na precisão do GPS sub-métrico (The influence of different vegetated canopy covers in sub-meter GPS accuracy). *In Simpósio Internacional de Iniciação Científica da Universidade de São Paulo. USP*.

Invited Conference Talks:

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| ▪ Australian Institute of Agricultural Science and Technology | 2017 |
| ▪ Tasmanian Dairy Conference | 2017 |
| ▪ Australian Dairy Conference | 2016 |

Press Appearances:

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| ▪ Material Talks (PanAnalytical blog)
<i>Determining Biophysical And Biochemical Attributes Of Pasture Using High Resolution Remote Sensing Techniques</i> | 2019 |
| ▪ The Advocate (Australian News)
<i>TIA researchers are studying precision forage management</i> | 2018 |
| ▪ Farming Ahead (Australian Magazine)
<i>Bird's eye view of pasture</i> | 2017 |